

Hallucination Forecasting in Multi-turn Conversation

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ABSTRACT Large language models (LLMs) are now widely deployed in multi-turn conversational systems, yet as conversations grow longer, models increasingly struggle to maintain consistent context and tend to generate unsupported or fabricated information. In this study, we frame hallucination forecasting in multi-turn dialogue as a task of predicting hallucination risk in future turns from dialogue history alone, before any target response is produced. This stands in contrast to post-hoc detection, which judges a response only after it has already been generated. Such a setting offers little practical value when the goal is prevention. Prior work has largely treated hallucination as a detection problem, leaving proactive risk estimation in live conversations largely unexplored. We build on the DiaHalu dataset and estimate hallucination risk in upcoming turns from token-level uncertainty and confidence signals. Across multiple GPT-family models, we find that these signals provide useful information for predicting future hallucination risk, and that contrary to the intuition that high uncertainty precedes hallucination low-entropy, high-confidence responses are more strongly associated with subsequent hallucination, a pattern we describe as an overconfident state preceding hallucination. Taken together, these findings suggest that hallucination risk in multi-turn dialogue can be estimated in advance, pointing toward the development of forecasting-based early-warning systems that move beyond reactive detection.

INDEX TERMS large language models, hallucination forecasting, multi-turn dialogue, uncertainty estimation, entropy, conversational AI

I. INTRODUCTION

Large language models (LLMs) are no longer used merely as tools for answering single questions, but have become core components of conversational systems that interact with users over multiple turns. In real-world usage, users do not simply provide a single self-contained question; instead, they ask follow-up questions based on previous responses, request additional information, and gradually specify the goal of the conversation. In such multi-turn conversation settings, it is important for a model not only to generate a response at each turn, but also to continuously track the previous dialogue context and produce answers that do not conflict with the accumulated information.

These characteristics of multi-turn conversation make the hallucination problem more complex. In a single-turn setting, the factuality or groundedness of a response can be evaluated relatively clearly. In a multi-turn dialogue, however, the context established by earlier utterances, shifts in the user's intent, and intermediate information generated by

the model itself can all influence later responses. During this process, the model may present unsupported information as factual, generate information that conflicts with previous dialogue content, or provide overly assertive answers under uncertainty. In particular, such hallucinations may not occur suddenly and independently in a specific response, but may instead relate to uncertainty dynamics or failures in context tracking in earlier turns.

A large body of prior work on hallucination has focused on detection, where the goal is to determine whether an already generated response contains hallucinated content. This approach is useful for evaluating the reliability of model outputs, but post-hoc judgment alone is insufficient for real-world conversational systems. If hallucination is detected only after the user has already received incorrect information, the system remains limited to responding after the error has occurred rather than preventing it in advance. Therefore, multi-turn dialogue systems require the ability not merely to determine whether the current response is hallucinated, but to predict

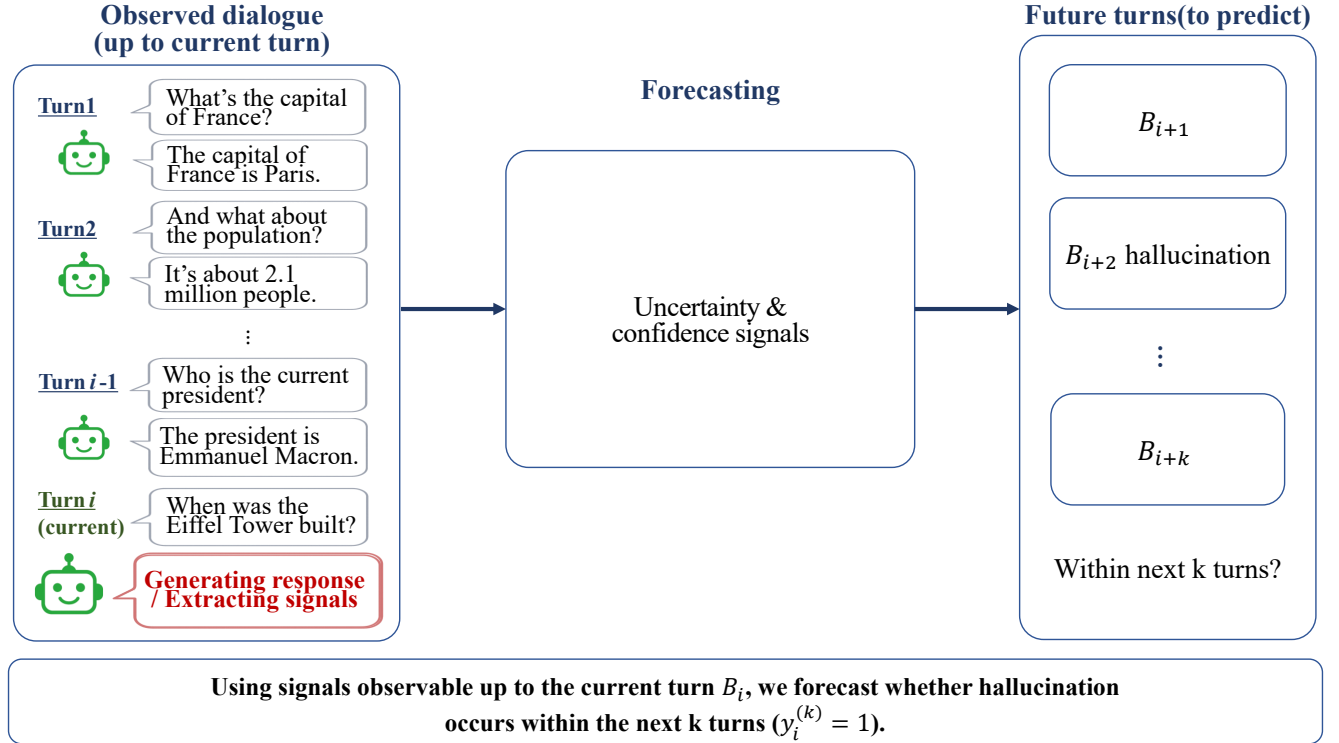


FIGURE 1: Overview of the proposed hallucination forecasting task. Given previous dialogue turns and their token-level uncertainty signals, the model predicts the likelihood of hallucination occurrence in upcoming turns.

the likelihood of future errors from the dialogue state before hallucination explicitly occurs.

Motivated by this problem, we redefine hallucination in multi-turn conversations as a future risk forecasting problem. Specifically, we aim to predict the likelihood that hallucination will occur in later turns based only on the dialogue history up to a given point, without using any information from after the target response has been generated. This setting differs from conventional turn-level hallucination detection. While detection determines whether a given response is hallucinated, forecasting estimates the risk of a future response that has not yet been generated from the dialogue trajectory observed so far.

If hallucination risk can be reliably estimated before a response is generated, it opens up a range of practical intervention strategies that post-hoc detection cannot support. A system aware of elevated risk at a given dialogue state could, for instance, prompt the model to seek clarification, switch to a more conservative generation strategy, or alert downstream components before an erroneous response reaches the user. In this sense, forecasting reframes hallucination mitigation as a proactive system-level capability rather than a retrospective quality check.

Figure 1 illustrates the overall framework of the proposed hallucination forecasting task.

To operationalize hallucination forecasting, we propose a framework that predicts future hallucination risk from dialogue history and token-level signals. For each dialogue turn,

we compute a range of token-level uncertainty and confidence signals, including entropy-based and log-probability-based signals, as well as response length as a simple baseline. In addition, we consider trajectory signals that capture how these quantities change across turns, to examine whether the temporal dynamics of uncertainty add predictive value beyond single-turn signals. These signals are used to forecast whether hallucination will occur in subsequent turns without observing the responses of those future turns. Through this formulation, we investigate whether token-level signals observable up to the current turn can serve as indicators of future hallucination in multi-turn dialogue.

To this end, we use the DiaHalu [1] dataset as a testbed for studying hallucination risk in multi-turn dialogue. For each dialogue turn, we construct a forecasting setting in which only the dialogue history available before the target response is used to predict whether hallucination will occur in the next or subsequent turns. This formulation differs from standard hallucination detection because it excludes the response to be judged and instead focuses on signals that are observable before the hallucination becomes explicit. By doing so, our setting better reflects practical early-warning scenarios in conversational systems, where the goal is to identify risky dialogue states before users are exposed to erroneous information.

The main contributions of this study are as follows:

- 1) We formulate hallucination forecasting in multi-turn conversations as a future risk prediction task, rather

than treating hallucination only as a post-hoc detection problem.

- 2) Using the DiaHalu dataset, we construct an experimental setting that predicts the likelihood of future hallucination at each dialogue turn.
- 3) We analyze whether token-level signals observable during response generation can serve as risk indicators before hallucination occurs.
- 4) We identify which token-level signals are most predictive of future hallucination, providing single-signal baselines and key empirical findings (e.g., the overconfident-state pattern) for future hallucination forecasting research.

II. RELATED WORK

This study is related to three research areas: performance degradation and context drift in multi-turn conversations, dialogue-level hallucination evaluation, and uncertainty-based hallucination analysis. Prior research has investigated the causes of hallucinations in large language models and proposed various methods to detect them. Moreover, recent work has actively examined the limitations of models in tracking context and their performance decline in multi-turn conversations. However, most of these studies have focused on assessing whether hallucinations exist in already generated responses or measuring the risk state at the current turn. In contrast, our study adopts a forecasting perspective that aims to predict the risk of hallucination in future turns. In this section, we review related work along four dimensions: (1) performance degradation and context drift in multi-turn settings, (2) dialogue-level hallucination benchmarks, (3) uncertainty-based hallucination detection, and (4) early warning signals and forecasting approaches.

A. PERFORMANCE DEGRADATION AND CONTEXT DRIFT IN MULTI-TURN CONVERSATIONS

Recent studies have reported that large language models can experience significant performance degradation in multi-turn conversational settings compared to single-turn interactions. Laban et al. [2] proposed an evaluation framework in which existing single-turn benchmarks are divided into multiple information shards and presented sequentially. Their experiments showed that performance substantially decreases when the same information is delivered sequentially in a multi-turn format, compared to providing all information in a single input. This decline cannot be explained solely by insufficient information or input length, and instead appears to stem from the multi-turn interaction itself. Additionally, models tend to make assumptions without sufficient evidence in early turns or rely excessively on previous incorrect responses, often failing to recover once the dialogue progresses in an incorrect direction.

To explain this performance degradation in multi-turn settings, research on context drift has also been conducted. Dongre et al. [3] quantified drift by measuring KL divergence between the token distributions of the test model and a

goal-aligned reference model. Although drift was previously assumed to accumulate continuously with dialogue length, results showed that it tends to reach an equilibrium at a certain level. Moreover, simple interventions such as goal reminders can effectively mitigate drift. These findings indicate that drift is not merely observable as a failure in the final output but represents a dynamic signal traceable throughout the dialogue.

DeepContext [4] utilized a stateful RNN architecture to detect semantic drift in multi-turn dialogues. This approach treats each turn's representation as a time series, continuously tracking the evolving dialogue state and capturing risk signals that may accumulate across multiple turns, which cannot be observed in individual responses alone. Notably, even seemingly harmless repeated responses can cause gradual bias in the model's internal state, potentially leading to inappropriate outputs later.

Recent work also highlights that multi-turn conversational settings can exacerbate hallucinations. HalluHard [5] proposed a benchmark for evaluating multi-turn hallucinations in high-risk domains such as law, medicine, and coding, demonstrating that hallucination frequency increases as the dialogue context grows. Similarly, ERGO [6] interprets entropy spikes during multi-turn dialogues as signals of conversational drift and proposes real-time monitoring to detect dialogue quality degradation. These studies collectively indicate that changes in model uncertainty and contextual state during dialogue progression are closely associated with subsequent errors.

B. DIALOGUE-LEVEL HALLUCINATION EVALUATION AND BENCHMARKS

To systematically analyze hallucinations in multi-turn dialogues, various benchmark datasets have been proposed. HalluDial [7] is a large-scale benchmark for dialogue-level hallucination evaluation, covering diverse conversational scenarios that include both spontaneous and induced hallucinations. It also incorporates both factuality-based and faithfulness-based hallucinations, allowing a comprehensive assessment of errors occurring throughout the dialogue.

DiaHalu is another dataset constructed to analyze hallucinations in multi-turn conversational settings, with each dialogue turn annotated for hallucination occurrences. This allows tracking hallucination patterns at the dialogue level. The dataset can be used not only for detection tasks but also for forecasting settings, where the goal is to predict the risk of future hallucinations based on preceding dialogue context.

However, most existing benchmarks focus primarily on evaluating whether generated responses contain hallucinations. The problem of forecasting future hallucination risk in multi-turn dialogues remains underexplored. This study leverages the DiaHalu dataset to establish a novel forecasting setting, predicting the likelihood of hallucinations occurring in upcoming turns at each dialogue point.

C. UNCERTAINTY-BASED HALLUCINATION DETECTION

Recent studies have demonstrated that internal uncertainty signals in large language models (LLMs) are closely asso-

ciated with hallucinations. Farquhar et al. [8] proposed semantic entropy, extending token-probability-based entropy to capture uncertainty at the semantic level. This allows measuring uncertainty across semantically equivalent responses, providing effective signals for hallucination detection.

HaluNet [9] leverages both token-level uncertainty and semantic representation uncertainty to detect hallucinations. This approach shows that uncertainty present in the internal representation space, beyond surface output probabilities, can serve as a critical cue for hallucination detection. INSIDE [10] also uses internal state information, including predictive confidence and entropy, to detect hallucinations, demonstrating that internal signals alone can support reliability assessments.

SelfCheckGPT [11] detects hallucinations by comparing consistency across multiple responses generated for the same prompt. Similarly, Entropy Production Rate (EPR) [12] uses token entropy changes in black-box settings to identify hallucinations. The Illusion of Progress [13] highlights limitations in previous hallucination detection studies, showing that simple statistical features such as response length can sometimes achieve performance comparable to semantic entropy.

Prior research has successfully leveraged various uncertainty signals including entropy, confidence, and consistency for hallucination detection. However, most approaches focus on detecting hallucinations in already generated outputs, and whether these uncertainty signals can be used to predict hallucinations in future turns remains largely unexplored.

D. PRE-RESPONSE RISK SIGNALS AND FORECASTING PERSPECTIVE

Recent studies have reported that risk signals can be observed not only after hallucinations occur, but also prior to response generation. Language Models (Mostly) Know What They Know [14] demonstrated that language models can partially distinguish between information they know and do not know, and that internal probability distributions include signals related to the likelihood of correct answers.

LLM Internal States Reveal Hallucination Risk Faced With a Query [15] experimentally showed that differences in the internal states of the model can appear before hallucinated or faithful responses are generated. In particular, it demonstrated that internal representations prior to response generation can predict the risk of hallucination, indicating the presence of observable early warning signals before errors manifest.

Enhancing Hallucination Detection via Future Context [16] proposed a method to leverage future context after hallucinations to improve detection performance. Although it focuses on detection, it highlights that hallucinations can persist and propagate across multiple turns, connecting this approach to the forecasting perspective.

Research in early anomaly detection also provides relevant insights. Early Anomaly Detection in Time Series [17] emphasizes the importance of detecting risks before the event occurs rather than relying solely on classification accuracy. This is directly relevant to hallucination forecasting, where

the goal is not only to evaluate the reliability of current responses but also to predict the risk of hallucination in future turns.

However, most prior work focuses on uncertainty analysis in single-turn settings or detection of hallucinations in the current response. Furthermore, even when analyzing pre-response risk signals, forecasting hallucinations in future turns is generally not addressed. To address this gap, we propose a forecasting framework that utilizes dialogue history and internal uncertainty signals in multi-turn conversations to predict the risk of future hallucinations. Unlike conventional detection settings that evaluate already generated responses, our forecasting setting predicts whether hallucinations will occur within the next k turns using only the dialogue history up to the current point.

III. PRELIMINARIES

In this section, we introduce the notation and problem setting used throughout this study. We first describe the structure of multi-turn dialogue and the token-level signals observable during response generation. We then define the hallucination forecasting task and present the evaluation metric used in our experiments.

A. MULTI-TURN DIALOGUE

A multi-turn dialogue consists of a sequence of alternating user utterances and assistant responses. We denote the user turn as A_i and the assistant response turn as B_i , and represent a dialogue as

$$D = (A_1, B_1, A_2, B_2, \dots, A_T, B_T), \quad (1)$$

where T denotes the number of assistant turns in the dialogue. Each assistant turn B_i is associated with a binary label $y_i \in \{0, 1\}$ indicating whether hallucination occurs at that turn. Specifically, $y_i = 1$ indicates that the response contains hallucination, whereas $y_i = 0$ indicates otherwise.

In multi-turn settings, contextual information accumulates as the conversation progresses, requiring the model to continuously track information from previous turns. Consequently, hallucinations may not arise independently within a single response, but may instead be associated with contextual information accumulated across multiple turns.

B. TOKEN-LEVEL SIGNALS

During response generation, large language models compute a probability distribution over candidate next tokens at each token position. These token-level probability distributions provide various signals that reflect the model's uncertainty and confidence.

One representative uncertainty signal is token entropy. Let $p_j(w)$ denote the probability assigned to candidate token w at the j -th token position. The entropy at that position is defined as

$$H(j) = - \sum_w p_j(w) \log p_j(w). \quad (2)$$

A higher entropy value indicates that the model distributes probability mass across multiple candidate tokens, reflecting greater uncertainty. Conversely, a lower entropy value indicates stronger confidence in a particular token.

In addition to entropy, various token-level signals can be derived, including mean log probability, probability margins between candidate tokens, vocabulary coverage, and response length. In this study, we investigate how these signals relate to future hallucination risk.

C. HALLUCINATION FORECASTING

Conventional hallucination detection aims to determine whether an already generated response B_i contains hallucinated content. In contrast, this study considers a forecasting setting, where the objective is to predict whether hallucination will occur in future turns based on the dialogue history observed so far.

To this end, we define a forecasting label that indicates whether hallucination occurs within the next k assistant turns after the current turn B_i :

$$y_i^{(k)} = \begin{cases} 1, & \text{if } \exists h \text{ such that } i < h \leq i + k \text{ and } y_h = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Thus, $y_i^{(k)} = 1$ indicates that hallucination occurs at least once within the next k turns following the current turn. The goal of this study is to predict this future hallucination risk using token-level signals observable at the current turn.

D. EVALUATION METRIC

We evaluate forecasting performance using the Area Under the Receiver Operating Characteristic Curve (AUROC). AUROC is a widely used metric that measures the ability of a signal to distinguish positive and negative instances without relying on a specific decision threshold.

An AUROC of 0.5 indicates random guessing. An AUROC greater than 0.5 indicates that larger signal values are associated with a higher likelihood of future hallucination, whereas an AUROC below 0.5 indicates that smaller signal values are associated with a higher likelihood of future hallucination.

By comparing the AUROC values of different token-level signals, we assess their effectiveness for forecasting future hallucination risk.

IV. PROPOSED METHOD

In this study, we formulate the task as forecasting the risk of hallucination in a multi-turn conversational setting, rather than detecting hallucinations retrospectively. We focus on token-level signals observable during the dialogue, analyzing whether these features can serve as early indicators of potential hallucinations. The overall framework consists of the following steps: 1) extraction of assistant responses from multi-turn dialogues, 2) computation of token-level log probability-based signals, 3) construction of forecasting features, 4) generation of k -window forecasting labels, and 5) prediction of future hallucination risk.

A. PROBLEM FORMULATION

We define a multi-turn conversation as

$$D = \{A_1, B_1, A_2, B_2, \dots, A_T, B_T\},$$

Traditional hallucination detection aims to determine whether a specific response B_i contains hallucinated content. In contrast, our study addresses a forecasting problem: using only the dialogue history up to the current turn to predict the likelihood of a hallucination occurring in future turns. Specifically, if a hallucination occurs at B_h , we assign a positive label to any prior assistant turn B_i that satisfies $i < h \leq i + k$. That is, the current turn is considered at risk if a hallucination occurs within the next k turns. This formulation allows us to predict whether the current dialogue state may lead to future hallucinations, rather than merely evaluating whether the current response is hallucinated.

B. POINT SIGNALS

For each assistant turn, we first extract point signals that can be computed independently. For each assistant response, we extract token-level log probabilities and compute the entropy of each token based on its probability distribution:

$$H(x) = - \sum_j p(x_j) \log p(x_j), \quad (4)$$

where $p(x_j)$ denotes the probability of token x_j . Since each assistant response consists of multiple tokens, we define the response-level entropy as the mean token entropy:

$$H_i = \frac{1}{N} \sum_{j=1}^N H(x_j), \quad (5)$$

where N is the number of tokens in the response and H_i represents the mean entropy for turn i . In addition to token entropy, we also compute various confidence-related point signals, such as the mean and minimum log probabilities of generated tokens, the difference between the top-1 and top-2 token probabilities, and the probability mass of top vocabulary candidates. Moreover, we include response length as a baseline feature, as recent hallucination detection studies have shown it to be a simple yet competitive signal [13].

C. TRAJECTORY SIGNALS

While point signals capture the state of an individual turn, multi-turn dialogues also contain temporal information that cannot be observed from a single turn alone. Since hallucinations may be associated with accumulated changes across previous turns, it is important to consider not only the current state of the model but also how its uncertainty and confidence evolve throughout the conversation. To capture such temporal dynamics, we construct trajectory signals based on changes across turns. Specifically, we compute the entropy delta between consecutive turns, the slope of the previous turn's entropy (prev-entropy slope), the moving average of entropy over recent turns (entropy rolling3), and the delta of log probabilities (logprob delta). Unlike point signals, which

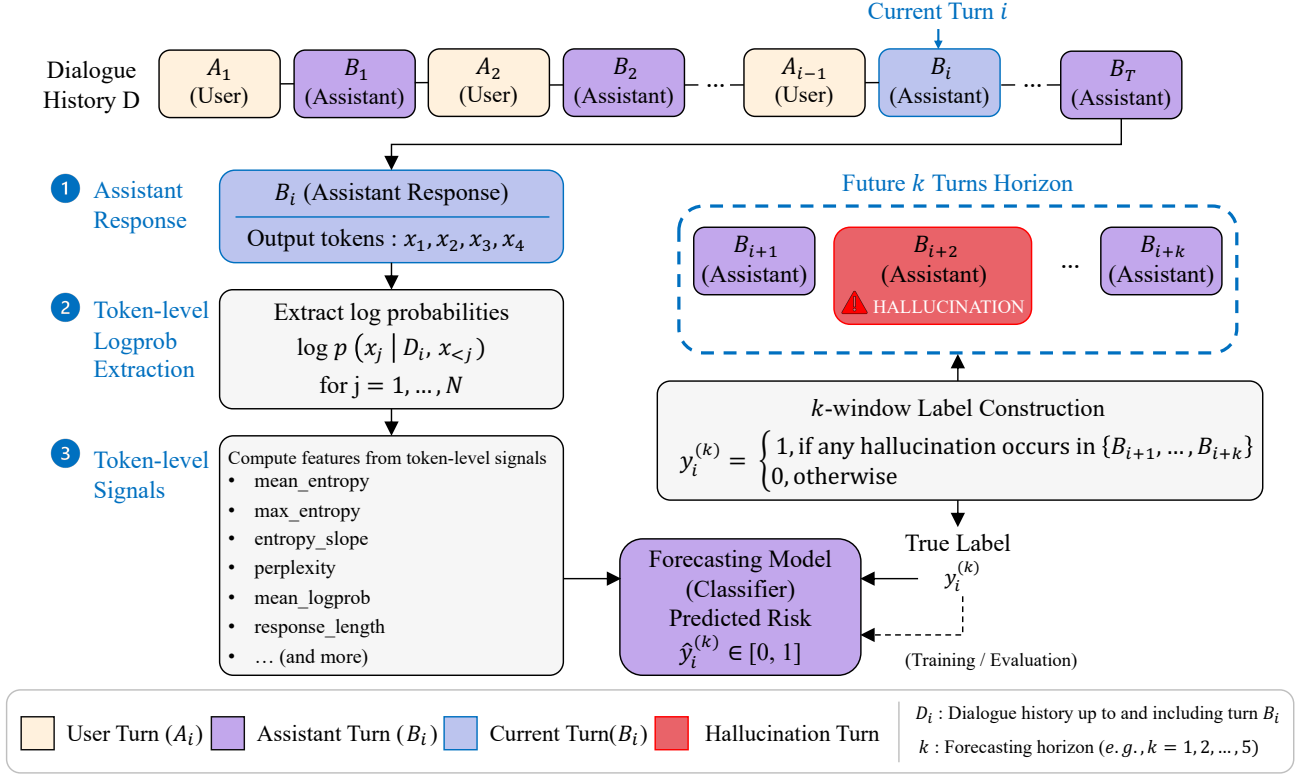


FIGURE 2: Overview of our proposed multi-turn hallucination forecasting framework. The diagram illustrates the extraction of token-level features from the current assistant turn, computation of entropy-based signals, construction of k -window forecasting labels, and prediction of future hallucination risk.

describe the model's state at a specific turn, trajectory signals explicitly incorporate information from multiple turns. These signals are intended to capture how the model's uncertainty and confidence evolve as the dialogue progresses, thereby enabling us to investigate whether temporal dynamics provide additional forecasting information beyond single-turn signals.

D. K-WINDOW HALLUCINATION FORECASTING

To construct the forecasting setting, a future hallucination label is generated for each assistant turn. Let B_h denote a turn in which a hallucination occurs. The forecasting label for the current turn B_i is defined as follows:

$$y_i^{(k)} = \begin{cases} 1 & \text{if } i < h \leq i + k \\ 0 & \text{otherwise} \end{cases}$$

That is, a turn is labeled positive if a hallucination occurs within the next k turns, and negative otherwise. For example, if a hallucination occurs at B_7 and $k = 2$, then B_5 and B_6 are labeled positive, while B_1 through B_4 are labeled negative. To analyze the impact of different forecasting horizons on forecasting performance, k is varied over the range 1 to 5. In addition to varying the horizon k , we also analyze how forecasting performance depends on the origin turn i at which

the prediction is made. This allows us to assess how the forecasting performance of token-level signals changes with both the temporal distance to the hallucination event and the position within the dialogue. This formulation defines a forecasting framework for predicting future hallucination risk.

E. OVERALL FRAMEWORK

We formulate hallucination forecasting as a binary classification problem: given the dialogue history up to the current assistant turn, we predict whether at least one hallucination occurs within the next k turns. The overall system operates according to the following sequence: 1) input of multi-turn dialogue, 2) extraction of assistant responses, 3) token-level log probability extraction, 4) computation of point and trajectory signals, 5) generation of k -window forecasting labels, and 6) prediction of future hallucination risk. By combining the token-level signals defined above with turn-level hallucination annotations, the framework enables forecasting of future hallucination risk. Rather than judging whether the current response is hallucinated, it leverages signals observable up to the current turn to predict the risk of hallucination in upcoming turns.

V. EXPERIMENTS

A. DATASET

This study uses the DiaHalu [1] dataset to evaluate hallucination forecasting in multi-turn dialogue. DiaHalu is a benchmark specifically designed for analyzing hallucinations in multi-turn conversational settings and provides annotations indicating whether hallucination occurs at the turn level.

Unlike conventional conversational QA datasets such as CoQA and QuAC, which do not explicitly annotate the locations of hallucinations and therefore require additional manual labeling to construct a forecasting setting, DiaHalu provides turn-level hallucination annotations. This makes it possible to formulate a future hallucination forecasting task more directly.

DiaHalu consists of 1,103 dialogues and 7,621 assistant turns. Each dialogue belongs to one of four domains: World Knowledge (371 dialogues), Chit-Chat (263), Task-oriented Style (210), and Reasoning (259). At the dialogue level, 476 dialogues contain at least one hallucination, while 627 dialogues contain none.

B. EXPERIMENTAL SETUP

We extract token-level log probabilities (logprobs) from assistant responses using the OpenRouter API. Among GPT-family models that provide logprobs, we select three models GPT-4o, GPT-4o-mini, and GPT-5.4 considering their release recency and model scale. By extracting signals from the same DiaHalu dialogues with all three models, we examine whether the observed signal patterns are not specific to a single model.

The generation parameters are set as follows. We fix the temperature to 0 to ensure reproducibility of the outputs. We set the maximum number of generated tokens to 300 to keep response lengths within a consistent range, which stabilizes signal computation and controls API cost. We set top-logprobs to 5 so that token-level signals such as entropy can be computed from the top candidate token probabilities returned by the API.

For each assistant turn B_i , the dialogue context up to that turn is provided as input, and the model generates a response from which token-level signals are extracted. We compute several entropy-based signals, including mean entropy (mean_entropy), maximum entropy (max_entropy), entropy standard deviation (entropy_std), and entropy slope across the response (entropy_slope). We also compute confidence-related signals, including perplexity, mean and minimum token log probability (mean_logprob and min_logprob), the probability gap between the top-1 and top-2 candidate tokens (mean_top1_minus_top2), and the probability mass covered by the top candidate tokens (mean_vocab_coverage). In addition, response_length is included as a simple baseline, following prior work showing that response length can perform competitively with more sophisticated uncertainty measures in hallucination detection [13]. Comparing our token-level signals against this baseline allows us to assess whether they capture information beyond a simple length heuristic.

The forecasting setting is constructed using the proposed k -window formulation. A turn is labeled as positive if at least one hallucination occurs within the next k assistant turns after the current turn B_i . To analyze the effect of forecasting horizon, experiments are conducted with $k \in \{1, 2, 3, 4, 5\}$. Turns that are already labeled as hallucinations are excluded from evaluation to avoid data leakage, since signals extracted from erroneous turns could otherwise be used to predict future errors. As a result, the forecasting evaluation set contains 7,132 turns for each model.

Performance is evaluated using AUROC, which measures discriminative ability independently of any specific classification threshold. Rather than training a separate classifier, we directly evaluate the standalone forecasting ability of each token-level signal using AUROC. We report two variants. Raw AUROC is our primary metric: it preserves the direction of each signal, so a value below 0.5 is itself an informative finding, indicating that lower signal values precede hallucination. Effective AUROC, defined as $\max(\text{AUROC}, 1 - \text{AUROC})$, abstracts away signal direction and is reported as a supplementary measure of overall discriminative strength. Unless otherwise noted, our conclusions about signal direction (e.g., the overconfident-state finding) are drawn from raw AUROC, while effective AUROC is used when comparing overall predictive strength across signals.

C. EXPERIMENTAL DESIGN

We design a forecasting framework by combining the token-level signals described above with the temporal annotations provided in the benchmark dataset.

The core task is to perform binary classification of whether at least one hallucination will occur within the next k turns from a given assistant turn. Forecasting labels are assigned according to the k -window formulation introduced earlier. The primary features used for prediction include entropy-based signals and confidence-based signals such as mean log probability, top1-top2 margin, and vocabulary coverage. Following prior work, response length is also included as a baseline feature. In addition, trajectory signals that capture turn-to-turn changes are included for comparison.

The overall experimental pipeline consists of four stages: (1) dialogue-context input and response generation, (2) extraction of token-level signals from token log probabilities, (3) generation of k -window forecasting labels, and (4) future hallucination prediction and evaluation.

The forecasting performance is evaluated using AUROC, which measures the ability to distinguish positive and negative examples without relying on a predefined threshold. Because AUROC is independent of feature scale, it provides an appropriate metric for objectively comparing the predictive power of heterogeneous token-level signals. Through this experimental design, we investigate whether model-internal signals observable during multi-turn dialogue can provide meaningful early indicators of future hallucination risk.

D. RESULTS

TABLE 1: Raw k_1 forecasting AUROC for each token-level signal. An AUROC of 0.5 corresponds to random guessing; the farther a value departs from 0.5 in either direction, the stronger its predictive association with future hallucination. Values above 0.5 indicate that higher signal values are associated with higher hallucination risk, whereas values below 0.5 (marked with an asterisk, *) indicate the inverse, i.e., lower signal values are associated with higher risk.

Signal	GPT-4o	GPT-4o-mini	GPT-5.4
mean_entropy	0.414*	0.402*	0.399*
max_entropy	0.508	0.502	0.449*
entropy_std	0.424*	0.416*	0.407*
entropy_slope	0.485*	0.448*	0.543
perplexity	0.416*	0.404*	0.394*
mean_logprob	0.584	0.596	0.606
min_logprob	0.500	0.506	0.544
mean_top1_minus_top2	0.586	0.597	0.594
mean_vocab_coverage	0.563	0.581	0.599
response_length	0.569	0.588	0.546

1) Signal AUROC Comparison

Table 1 presents the raw AUROC values of each signal for the k_1 forecasting task. The values can be interpreted as follows: an AUROC of 0.5 indicates no predictive association, while values farther from 0.5 in either direction indicate a stronger association with future hallucination. Values above 0.5 indicate that higher signal values are associated with higher hallucination risk, whereas values below 0.5 indicate the opposite. Among the token-level signals examined, confidence-related signals and entropy-based signals exhibited contrasting behaviors.

Specifically, confidence-related signals such as `mean_logprob`, `mean_top1_minus_top2`, and `mean_vocab_coverage` consistently achieved AUROC values above 0.5 across all three models. For example, `mean_top1_minus_top2` achieved AUROC scores of 0.586, 0.597, and 0.594 for GPT-4o, GPT-4o-mini, and GPT-5.4, respectively.

In contrast, entropy-based signals and perplexity consistently produced AUROC values below 0.5 across all models. For instance, `mean_entropy` achieved AUROC values of 0.414, 0.402, and 0.399 for GPT-4o, GPT-4o-mini, and GPT-5.4, respectively. Similarly, `perplexity` obtained AUROC values of 0.416, 0.404, and 0.394.

These results suggest that the common expectation that higher uncertainty leads to a greater risk of future hallucination is insufficient to explain the observed patterns. Instead, confidence-related signals appear to provide stronger information for forecasting future hallucinations.

2) Confidence-Related Signals and Future Hallucination Risk

To better understand the directionality of the observed signals, we compared the distributions of signals between cases where hallucination occurs in the next turn ($k_1 = 1$) and cases where it does not ($k_1 = 0$) (Fig. 3).

The analysis revealed that turns immediately preceding future hallucinations tend to exhibit lower `mean_entropy`,

while `mean_logprob` and `mean_top1_minus_top2` tend to be higher. In other words, future hallucination risk appears to be associated not with higher uncertainty, but rather with responses generated with higher confidence.

These findings suggest that hallucinations do not necessarily arise only under highly uncertain conditions; rather, they may also follow responses generated with strong token-level confidence. We interpret this phenomenon as an *overconfident state preceding hallucination*.

3) Point vs. Trajectory Signals

The analyses above focused on point signals that are computed independently for each turn. However, hallucinations may also be related to the accumulation of changes across previous turns. To investigate this possibility, we compared trajectory signals (`entropy_delta`, `prev_entropy_slope`, `entropy_rolling3`, and `logprob_delta`) with point signals using the same evaluation metric (Table 2). Because trajectory signals require information from preceding turns, this comparison is restricted to turns after B_2 , where such information is available. Point signals consistently achieved effective AUROC values in the range of 0.56–0.59 across all models. In contrast, trajectory signals directly based on turn-to-turn changes, such as `entropy_delta` and `logprob_delta`, remained close to chance level, with effective AUROC values around 0.51. The slope-based signal (`prev_entropy_slope`) showed only weak predictive ability, with effective AUROC values between 0.53 and 0.55. The rolling-average signal (`entropy_rolling3`) achieved performance comparable to that of point signals, ranging from 0.56 to 0.58.

Overall, trajectory signals did not provide additional predictive power beyond that of point signals. A more detailed interpretation of why turn-to-turn changes fail to capture useful predictive information is provided in the Discussion.

4) Scale Robustness

To determine whether the observed signal patterns were specific to a particular model, we compared the results across GPT-4o, GPT-4o-mini, and GPT-5.4 (Fig. 4).

Similar trends were observed across all three models. In particular, `mean_entropy` consistently produced AUROC values below 0.5 (0.414, 0.402, and 0.399, respectively), whereas confidence-related signals such as `mean_logprob`, `mean_top1_minus_top2`, and `mean_vocab_coverage` consistently achieved AUROC values above 0.5.

Although minor numerical differences existed across model sizes, the directionality of the signals and their relative rankings remained stable. This consistency indicates that the observed forecasting patterns generalize across different GPT-family models.

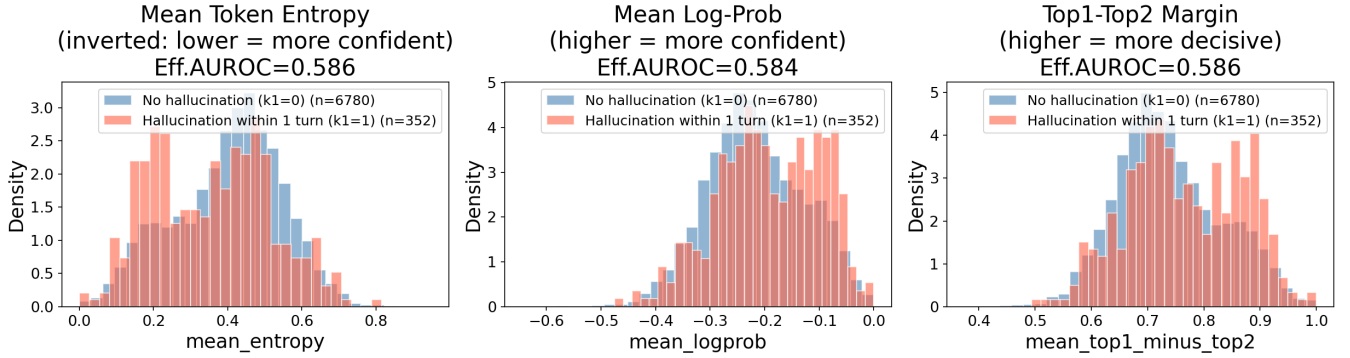
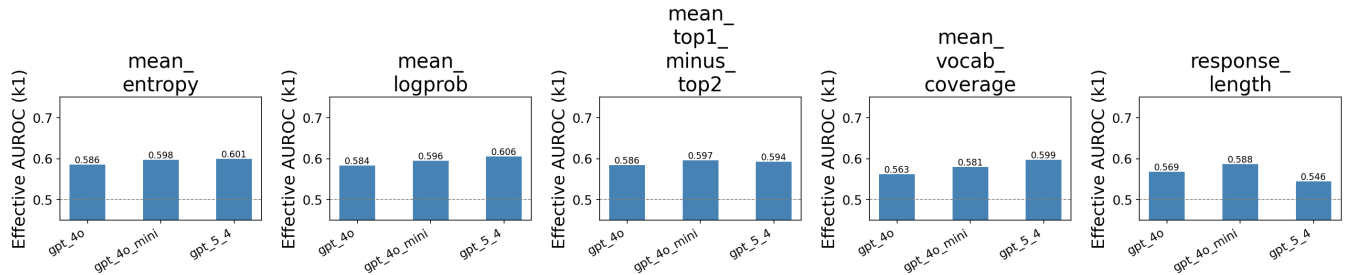
5) Domain Analysis

Fig. 5 illustrates the forecasting behavior of `mean_entropy` across different domains.

TABLE 2: Effective AUROC comparison between point and trajectory signals (k_1 , turns after B_2).

Signal	Type	GPT-4o	GPT-4o-mini	GPT-5.4
mean_entropy	point	0.563	0.581	0.580
mean_logprob	point	0.563	0.582	0.592
mean_top1_minus_top2	point	0.562	0.581	0.574
entropy_delta	trajectory	0.516	0.507	0.510
prev_entropy_slope	trajectory	0.531	0.550	0.533
logprob_delta	trajectory	0.509	0.507	0.518

Overconfident Hallucination: High-confidence turns precede more hallucinations

FIGURE 3: Signal distributions for turns followed by hallucination within one turn ($k_1 = 1$) versus those that are not ($k_1 = 0$). Turns preceding hallucination tend to show lower mean entropy and higher confidence (mean log-probability and top1–top2 margin), illustrating the overconfident state preceding hallucination.Signal Effective AUROC across GPT Model Scales (k_1)FIGURE 4: k_1 forecasting AUROC of representative signals across GPT-4o, GPT-4o-mini, and GPT-5.4. The direction and relative ranking of signals remain stable across models; entropy-based signals stay below 0.5 (shown in red) while confidence-related signals stay above 0.5.

The inverse relationship between entropy and future hallucination risk was particularly pronounced in the World Knowledge and Reasoning domains. In contrast, the effect was weaker in the Task-oriented Style domain, where AUROC values were closer to 0.5.

These findings suggest that the relationship between uncertainty and hallucination risk may vary across domains. However, because the number of samples differs across domains, these observations should be interpreted as preliminary.

6) Forecasting Horizon ($k_1 \sim k_5$)

We further examined how predictive performance changes as the forecasting horizon increases (Fig. 6).

Overall, the observed signal patterns remained relatively

stable from k_1 to k_5 . In particular, the inverse relationship of entropy-based signals and the predictive power of confidence-related signals were consistently observed for shorter horizons (k_1 – k_3).

As the forecasting horizon increased, the performance of simple statistical features such as `response_length` tended to improve relative to entropy-based signals. In some cases, `response_length` achieved performance comparable to or better than entropy-based signals.

These results suggest that token-level signals may provide predictive information across multiple forecasting horizons. At the same time, they highlight the importance of comparing uncertainty-based features against simple yet competitive

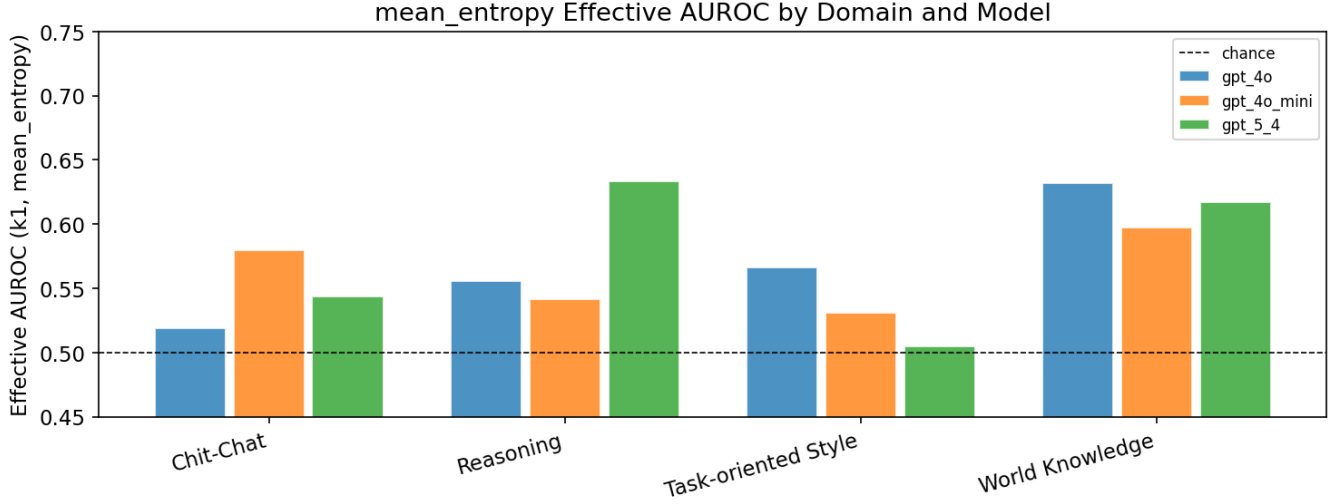


FIGURE 5: Effective k_1 AUROC of `mean_entropy` by domain and model. The inverse relationship between entropy and future hallucination risk is more pronounced in the World Knowledge and Reasoning domains, and weaker in the Task-oriented Style domain. Domain sample sizes are unequal, so these results are preliminary.

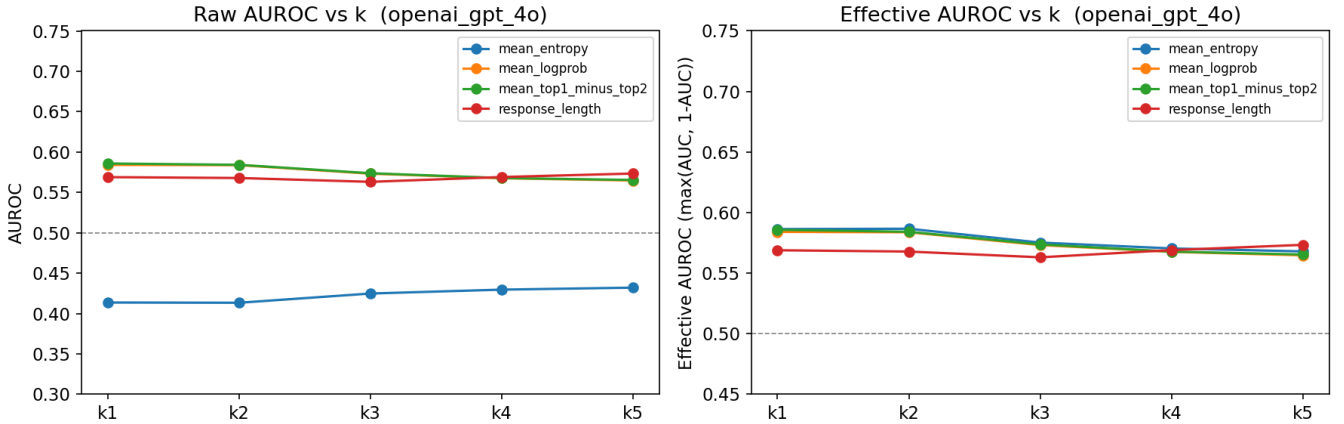


FIGURE 6: Raw and effective AUROC of representative signals as the forecasting horizon k increases from k_1 to k_5 (GPT-4o). The direction and relative ordering of signals remain stable across horizons.

baselines.

7) Forecasting by Origin Turn

Beyond varying the forecasting horizon k , we also analyze how forecasting behavior depends on the origin turn i from which the prediction is made (Fig. 7). This addresses whether the dialogue position itself affects both the model’s uncertainty and the predictability of future hallucination.

Two broad patterns emerge. First, mean entropy increases steadily as the dialogue progresses: for GPT-4o, the average `mean_entropy` rises from 0.341 at the first turn to roughly 0.45 in later turns. This is consistent across all three models and suggests that uncertainty accumulates as conversational context grows. Second, the forecasting strength of `mean_entropy` is not uniform across turns. It is relatively high at early origins (effective AUROC 0.593 at $i = 1$ and

0.649 at $i = 2$ for GPT-4o), drops to near-chance in the middle of the dialogue (around 0.50–0.52 for $i = 3$ to 6), and fluctuates at later turns.

We note that the number of forecasting instances decreases sharply at later turns (from over 1,000 at early turns to fewer than 150 by $i = 9$ for GPT-4o), so the high AUROC values at large i are based on limited data and should be interpreted with caution. Table 3 shows the results when varying both the origin turn i and the horizon k , confirming the same trend. Overall, this turn-level analysis indicates that entropy accumulates over the course of a dialogue, while the predictive value of token-level signals is strongest at the earlier forecasting origins.

8) Summary of Findings

The experimental results lead to five main observations.

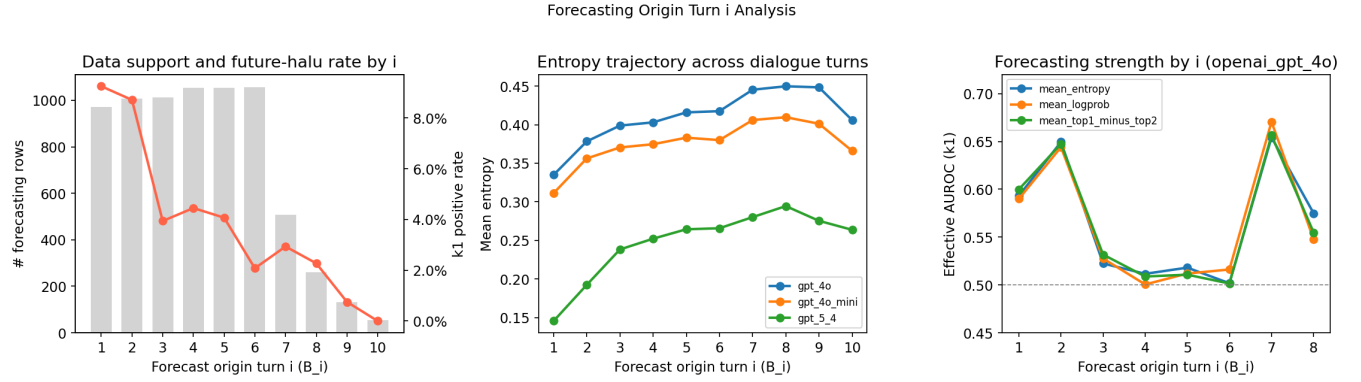


FIGURE 7: Forecasting behavior by origin turn i . Left: number of forecasting instances and k_1 positive rate by turn. Center: mean entropy rises as the dialogue progresses across all three models. Right: effective k_1 AUROC of point signals by origin turn (GPT-4o).

TABLE 3: Effective AUROC of `mean_entropy` across origin turn i (B_i) and forecasting horizon k for the three models. An asterisk (*) marks inverted raw AUROC, where lower entropy precedes future hallucination. Values at later turns are based on fewer forecasting instances and should be interpreted with caution.

Model	Origin	k_1	k_2	k_3	k_4	k_5
GPT-4o	B_1	0.593*	0.620*	0.596*	0.573*	0.555*
	B_2	0.649*	0.620*	0.592*	0.568*	0.562*
	B_3	0.522	0.516	0.509	0.509	0.506
	B_4	0.512	0.550	0.555	0.544	0.550
	B_5	0.518	0.512	0.506*	0.505*	0.501*
	B_6	0.502	0.559*	0.543*	0.544*	0.544*
	B_7	0.654*	0.600*	0.588*	0.588*	0.588*
	B_8	0.575*	0.575*	0.575*	0.575*	0.575*
GPT-4o-mini	B_1	0.590*	0.615*	0.595*	0.582*	0.561*
	B_2	0.674*	0.648*	0.618*	0.584*	0.577*
	B_3	0.535*	0.524*	0.500*	0.508*	0.515*
	B_4	0.516	0.526	0.507	0.506*	0.501*
	B_5	0.522	0.514	0.504*	0.509*	0.503*
	B_6	0.548*	0.568*	0.550*	0.548*	0.548*
	B_7	0.693*	0.643*	0.645*	0.645*	0.645*
	B_8	0.634*	0.554*	0.554*	0.554*	0.554*
GPT-5.4	B_1	0.577*	0.620*	0.611*	0.602*	0.579*
	B_2	0.688*	0.660*	0.627*	0.605*	0.593*
	B_3	0.580*	0.533*	0.517*	0.510*	0.507*
	B_4	0.566	0.587	0.603	0.607	0.611
	B_5	0.523	0.524	0.545	0.541	0.544
	B_6	0.524*	0.522*	0.508*	0.501*	0.501*
	B_7	0.514	0.517	0.520	0.520	0.520
	B_8	0.647*	0.625*	0.625*	0.625*	0.625*

First, token-level signals extracted during response generation provide useful information for forecasting future hallucination risk in multi-turn dialogue.

Second, contrary to conventional intuition, entropy-based signals often exhibit an inverse relationship with future hallucination risk, whereas confidence-related signals such as `mean_logprob` and `mean_top1_minus_top2` demonstrate stronger predictive power.

Third, trajectory signals that capture turn-to-turn changes do not provide additional predictive value beyond point signals, suggesting that hallucination risk is more closely associated with the model's confidence state at the current turn than with turn-to-turn changes.

Fourth, the analysis by origin turn shows that mean entropy accumulates as the dialogue progresses, while the forecasting performance of token-level signals varies with turn position and is strongest at earlier forecasting origins.

Finally, these patterns are observed consistently across different GPT-family models and forecasting horizons, indicating that these forecasting signals may generalize beyond a specific model.

VI. DISCUSSION

Our experiments provide evidence that token-level signals observable at the time of response generation carry meaningful predictive information about future hallucination risk in

multi-turn dialogue. Two observations stand out. First, across three GPT-family models, hallucination risk tends to increase following responses generated with higher confidence rather than higher uncertainty. Second, signals capturing turn-to-turn changes do not improve forecasting over single-turn signals. This section discusses the underlying reasons for these findings and their broader implications.

Why an Overconfident State Precedes Hallucination.

The most striking observation in our experiments is that entropy-based signals yielded AUROC values below 0.5 across all three models, whereas confidence-related signals such as `mean_logprob` and `mean_top1_minus_top2` showed stronger predictive power. This contradicts the common intuition that hallucination arises when the model is uncertain. We term this phenomenon an *overconfident state preceding hallucination*.

One possible mechanism is miscalibration. When a model generates fabricated content, it may not internally represent the content as uncertain, and instead assign high probability to the generated tokens. This aligns with recent empirical evidence by Miao and Kearns [18], who showed that LLMs frequently exhibit severe miscalibration and high surface-level confidence when producing fabricated information. Under this view, high token-level confidence reflects the model's internal certainty about its own generation, not the factual correctness of the content; when this certainty is misplaced, an overconfident response can precede subsequent errors. Consistent with this, our finding that entropy-based metrics fail to anticipate such failures echoes Ma et al. [19], who noted that standard probabilistic entropy metrics struggle when the model is confidently incorrect.

We emphasize that this interpretation is based on observed correlations; we do not directly establish a causal relationship between confidence and hallucination, and we leave its experimental verification to future work.

Hallucination Risk Appears More Related to Confidence State Than Temporal Change. Another key observation is that trajectory signals capturing turn-to-turn changes did not provide predictive power beyond single-turn point signals. In particular, signals that directly reflect the direction of change, such as `entropy_delta` and `logprob_delta`, remained close to chance level. This suggests that hallucination risk is better explained by “how confident the model is at the current turn” than by “how confidence changes over turns.” The fact that `entropy_rolling3`, a moving average over recent turns, performed comparably to point signals can likewise be explained by its tendency to reflect the current confidence level.

One caveat is that the dialogues in our dataset are relatively short on average, which limits the number of turns over which trajectory signals can be computed. The weak performance of trajectory signals may therefore be partly attributable to insufficient turn history rather than to a fundamental absence of temporal signal, and longer dialogues might reveal different dynamics.

Nevertheless, taken at face value, this observation has a

practical implication. If hallucination risk is primarily associated with the confidence state at the current turn, complex sequential models that model the entire turn history (e.g., RNN-based approaches) are unlikely to substantially outperform simple per-turn confidence thresholding. These findings suggest that forecasting may not necessarily require complex temporal models, and that signals observable at the current turn already contain meaningful predictive information.

Turn Position Shapes Both Uncertainty and Forecasting Performance. Our turn-level analysis offers further insight into how the position within a dialogue affects both uncertainty and forecasting. Mean entropy accumulates steadily as the dialogue progresses, consistent with the idea that a growing context introduces more competing information and raises the model's per-token uncertainty. At the same time, the forecasting performance of token-level signals is strongest at the earliest origin turns and weakens toward the middle of the dialogue. One possible interpretation is that early turns establish the factual grounding of the conversation, so the model's confidence at these turns is especially informative about whether the dialogue will eventually contain hallucination.

This pattern, however, must be interpreted in light of the dataset structure. The number of forecasting instances drops sharply at later turns, because most DiaHalu dialogues are relatively short, so only a small fraction of conversations reach high turn indices. As a result, the AUROC values at large i are estimated from limited samples and are correspondingly noisy, and the apparent rebounds in forecasting performance at very late turns should not be over-interpreted. This reinforces our earlier observation that the short average length of the dataset constrains turn-position and trajectory-based analyses, and suggests that the relationship between turn position and forecasting performance should be re-examined using datasets with longer and more balanced dialogues.

Limitations. This study has several limitations. First, the effective AUROC, which accounts for signal direction reversal, is a somewhat optimistic estimate because the direction is determined on the full dataset. We therefore report raw AUROC as the primary result and use effective AUROC only as an auxiliary indicator of the strength of direction reversal; a more rigorous evaluation would require validation through a train/test split. Second, because the number of samples differs across domains, the domain-level results should be interpreted as preliminary. Third, our experiments and findings are based on a single benchmark (DiaHalu) and have not been validated across multiple datasets; the generality of the observed patterns therefore remains to be established. Fourth, this study evaluates the standalone AUROC of each signal without training a separate classifier; the performance of learning-based predictors that combine multiple signals is left for future work. Fifth, our analysis is limited to GPT-family models, and further validation is needed to determine whether the observed patterns also hold for other open-source or non-GPT models. Sixth, because hallucination forecasting in multi-turn dialogue is a newly introduced task, there is no

established prior method that can serve as a direct comparison baseline. We therefore compare token-level signals against a simple response-length baseline rather than against existing detection systems, which are designed for a different task. Benchmarking against detection methods adapted for forecasting remains an important direction for future work.

Implications and Future Work. Our results demonstrate the potential of token-level confidence signals, observable at the time of response generation, to serve as early indicators of future hallucination risk. This provides initial evidence for extending hallucination mitigation from post-hoc detection toward proactive forecasting.

A central direction for future work is to test the causal mechanism suggested above: whether the miscalibration underlying overconfident responses can be directly linked to subsequent hallucination, for example through calibration-based analyses or controlled interventions on the model's confidence.

A second direction is to improve forecasting performance itself. In this study, we evaluated the standalone AUROC of each signal without training a classifier; combining multiple token-level signals, incorporating turn-position information (as suggested by our origin-turn analysis), and training learning-based predictors could yield stronger forecasting than any single signal alone.

A third direction is to use these forecasting signals to mitigate hallucination in practice. Once a turn is predicted to carry high future risk, a system could trigger mitigation strategies such as switching to a more conservative decoding strategy, prompting the model to verify uncertain claims, retrieving external evidence, or alerting the user before an erroneous response is delivered. Whether such forecasting-driven interventions measurably lower hallucination rates is an important question for future work.

Finally, extending the analysis to longer dialogues, a wider range of datasets, and other model families would allow a broader validation of the generality of the phenomenon of an overconfident state preceding hallucination, as well as a re-examination of whether temporal signals become informative when richer turn histories are available.

VII. CONCLUSION

In this paper, we reframed hallucination in multi-turn dialogue as a forecasting problem, predicting the risk of future hallucination from dialogue history available before a target response is generated. Using the DiaHalu dataset, we evaluated a range of token-level uncertainty and confidence signals across three GPT-family models (GPT-4o, GPT-4o-mini, and GPT-5.4). Our experiments revealed an overconfident state preceding hallucination, in which low-entropy, high-confidence responses are more strongly associated with subsequent hallucinations than uncertain ones, contrary to the common intuition that high uncertainty precedes errors. The consistency of this pattern across all three GPT-family models suggests that the observed relationship is robust across model variants, while its persistence across multiple forecasting

horizons indicates that it reflects a stable predictive pattern rather than a model-specific artifact. Overall, our findings suggest that token-level confidence signals are associated with future hallucination risk and may provide useful information for forecasting-based risk estimation in multi-turn dialogue systems.

VIII. ACKNOWLEDGEMENT

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) [RS-2021-II211341, Artificial Intelligence Graduate School Program (Chung-Ang University) and RS-2022-II220124, Development of Artificial Intelligence Technology for Self-Improving Competency-Aware Learning Capabilities]. We also acknowledge the use of the DiaHalu dataset for conducting the multi-turn hallucination forecasting experiments and the OpenRouter API for facilitating token-level log probability extraction across GPT-family models (GPT-4o, GPT-4o-mini, and GPT-5.4). We thank our colleagues for their valuable discussions and contributions to the experimental design, feature extraction methodology, and overall framework development. Their input was instrumental in the successful completion of this study.

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